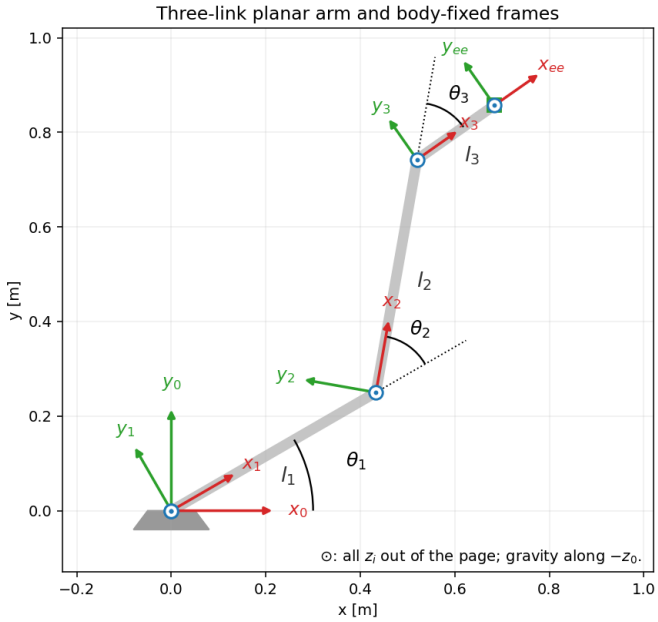
ECE276C Assignment 1

Robot Kinematics and Simple Control

Sachin Bhadang
PID: A69040722
Email: sbhadang@ucsd.edu

1. Arm and body-fixed coordinate frames



Frame $\{0\}$ is fixed at the base. Frame $\{i\}$ is attached to link i at joint i, with x_i along the link and z_i along the joint axis. Frame $\{ee\}$ is at the tip, with the same orientation as $\{3\}$. All frames are right-handed. The offsets follow assignment Table 1: $a_0 = 0$, $a_1 = l_1$, $a_2 = l_2$, $a_3 = l_3$.

2. End-effector kinematics

Let $\mathbf{q} = [\theta_1, \theta_2, \theta_3]^\top$ and $\mathbf{x} = f(\mathbf{q}) = [X, Y, \theta]^\top$. Then

$$\begin{aligned} X &= l_1 \cos\theta_1 + l_2 \cos(\theta_1 + \theta_2) + l_3 \cos(\theta_1 + \theta_2 + \theta_3), \\ Y &= l_1 \sin\theta_1 + l_2 \sin(\theta_1 + \theta_2) + l_3 \sin(\theta_1 + \theta_2 + \theta_3), \\ \theta &= \theta_1 + \theta_2 + \theta_3 \pmod{2\pi}. \end{aligned}$$

At $\mathbf{q} = \mathbf{0}$, the arm points along $+x_0$ and the tip is at $(l_1 + l_2 + l_3, 0)$.

3. Required PyBullet configurations

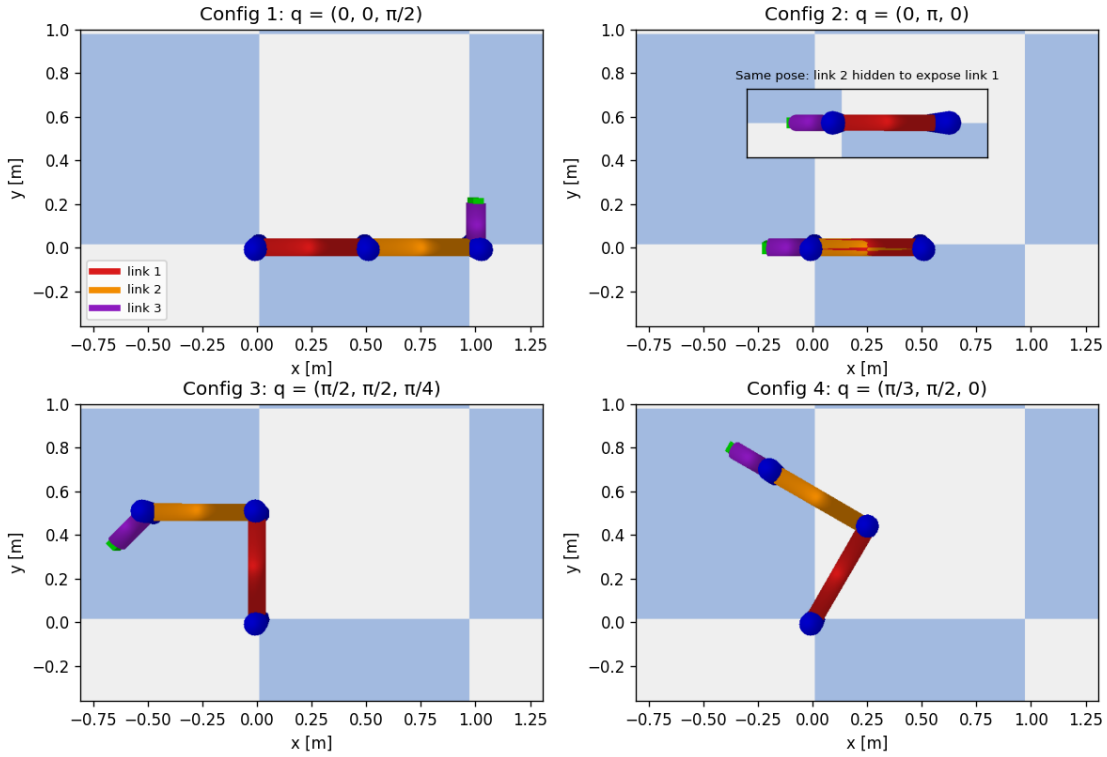
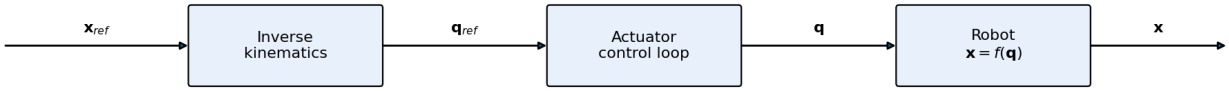


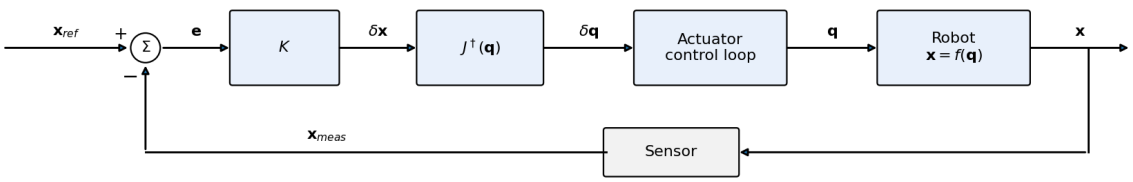
Figure 1: The supplied URDF with $l_1 = l_2 = 0.5\,\text{m}$ and $l_3 = 0.2\,\text{m}$. Links 1, 2 and 3 are red, orange and purple. In configuration 2, links 1 and 2 coincide exactly; the labeled inset hides link 2 to expose link 1 without changing the joint configuration.

4. Open-loop and closed-loop control diagrams

(a) Open-loop setpoint control



(b) Closed-loop setpoint control



Both diagrams use $\mathbf{x}_{ref} = [X, Y, \theta]^{\top}_{ref}$. In (b), $J = \partial f / \partial \mathbf{q} \in \mathbb{R}^{3\times 3}$, $\mathbf{e} = \mathbf{x}_{ref} - \mathbf{x}_{meas}$, and the orientation error is wrapped to $[-\pi, \pi)$.

5. Smooth profile and open-loop motion

For a move of duration T, let $\tau = t/T$ and use

$$\mathbf{x}_{ref}(t) = \mathbf{x}_0 + s(\tau)(\mathbf{x}_f - \mathbf{x}_0), \qquad s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5.$$

The polynomial satisfies the boundary conditions

$$s(0) = 0, \quad s(1) = 1, \quad s'(0) = s'(1) = 0, \quad s''(0) = s''(1) = 0,$$

where primes denote derivatives with respect to τ. In physical time,

$$\dot{s} = \frac{30\tau^2(1-\tau)^2}{T}, \qquad \ddot{s} = \frac{60\tau(1-\tau)(1-2\tau)}{T^2}.$$

The profile is second-order smooth (C^2): position, velocity and acceleration are continuous, including at the waypoints. All coordinates use the same s, so each Cartesian segment is straight.

The four moves $1 \to 2 \to 3 \to 4 \to 1$ take 2.5 s each, totaling 10 s; $\theta_{ref} = 0$ throughout. At 240 Hz, analytic inverse kinematics converts each sampled pose into joint setpoints on one continuous elbow branch. These drive PyBullet's position motors, with joint velocity feedforward; measured end-effector pose is used only for recording.

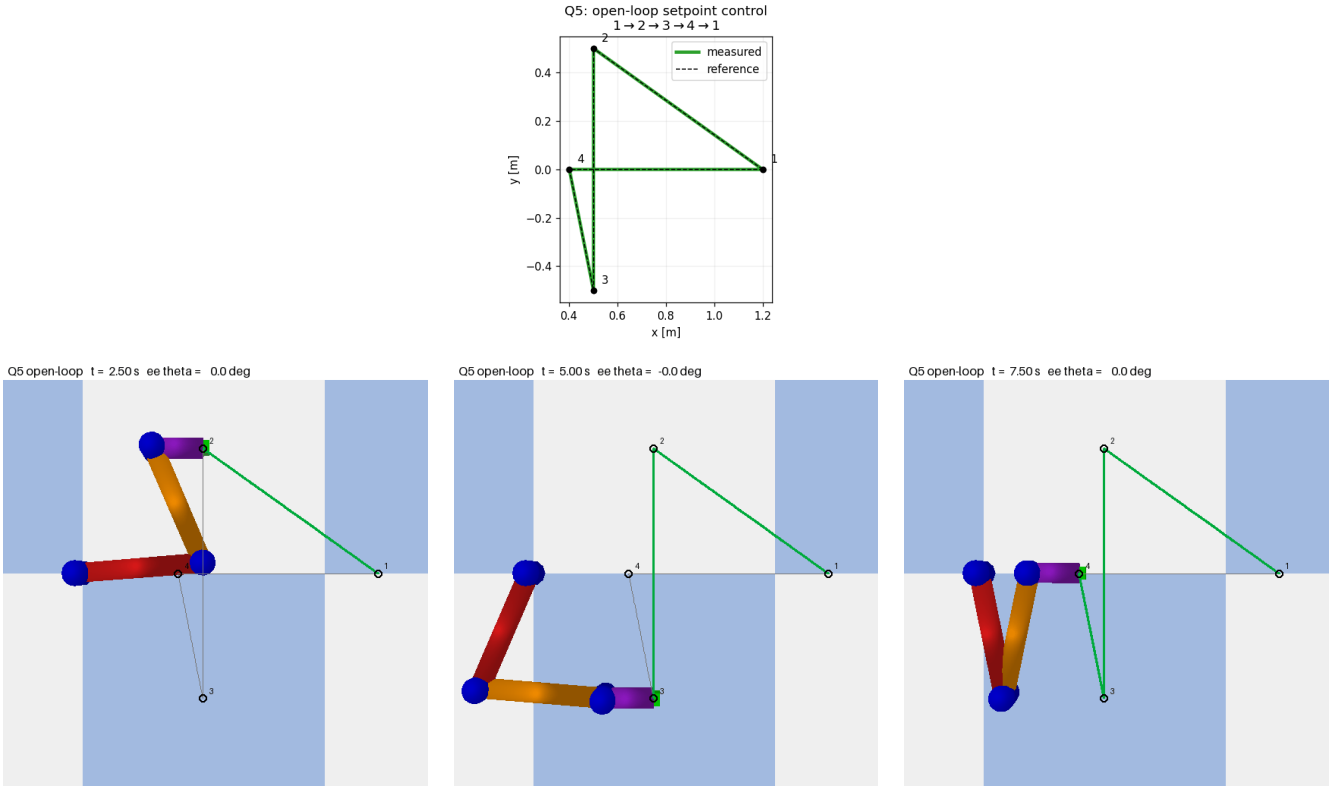


Figure 2: Q5: measured end-effector path (top) and animation frames at $t = 2.5$, 5.0 and 7.5 s (bottom, left to right), reaching locations 2, 3 and 4 with $\theta = 0$.

Animation: `q5_open_loop.gif` (20 fps, 10.05 s including the final frame). The reference velocity is zero at every waypoint. In the sampled simulation, waypoint speeds are below 6×10^{-5} m/s and the maximum position error is 3.9 μm.

6. Closed-loop motion and nullspace control

The task is now $\mathbf{x}_{xy} = [X, Y]^\top$; θ is free. Use the same reference path and timing as Q5, with $J_{xy} = \partial \mathbf{x}_{xy}/\partial \mathbf{q} \in \mathbb{R}^{2\times 3}$. Compute the SVD pseudoinverse $J_{xy}^\dagger$ and

$$N = I_3 - J_{xy}^\dagger J_{xy}, \qquad J_{xy} N = 0.$$

Define the interior angles at the shared joints by

$$\alpha_{12} = \pi - |\bar{\theta}_2|, \qquad \alpha_{23} = \pi - |\bar{\theta}_3|, \qquad \bar{\theta}_i = \mathrm{wrap}_{[-\pi,\pi)}(\theta_i).$$

A straight pair of links has the largest interior angle, π. To drive both angles toward this value subject to the position constraint, minimize their squared departures from π:

$$U(\mathbf{q}) = \tfrac{1}{2}\left[(\pi - \alpha_{12})^2 + (\pi - \alpha_{23})^2\right] = \tfrac{1}{2}(\bar{\theta}_2^2 + \bar{\theta}_3^2), \qquad \nabla_{\mathbf{q}} U = [0, \bar{\theta}_2, \bar{\theta}_3]^\top.$$

The angles stay away from the wrap discontinuity during this motion. Use projected gradient descent:

$$\dot{\mathbf{q}}_2 = -\lambda \nabla_{\mathbf{q}} U, \qquad \dot{\mathbf{q}}_{null} = N \dot{\mathbf{q}}_2, \qquad \lambda = 10\,\mathrm{s}^{-1}.$$

Combining it with the next-sample closed-loop trajectory step gives

$$\delta \mathbf{q} = J_{xy}^\dagger \big(\mathbf{x}_{xy,ref}[k+1] - \mathbf{x}_{xy,meas}[k]\big) + \Delta t\, \dot{\mathbf{q}}_{null}, \qquad \Delta t = 1/240\,\mathrm{s}.$$

Thus $J_{xy}\dot{\mathbf{q}}_{null} = 0$: the secondary controller changes posture without adding Cartesian velocity. This is a local optimization of U along the position constraint.

The implementation reads the joint angles and end-effector position each step. It corrects the finite-step linearization error with repeated Jacobian position corrections, then sends the resulting joint increment divided by Δt to the velocity motors. At location 1 the arm is straight and J_{xy} is singular; analytic IK selects the first nonzero departure sample, and the final sample returns to the straight posture. The simulated initial state is exactly $\mathbf{q} = \mathbf{0}$.

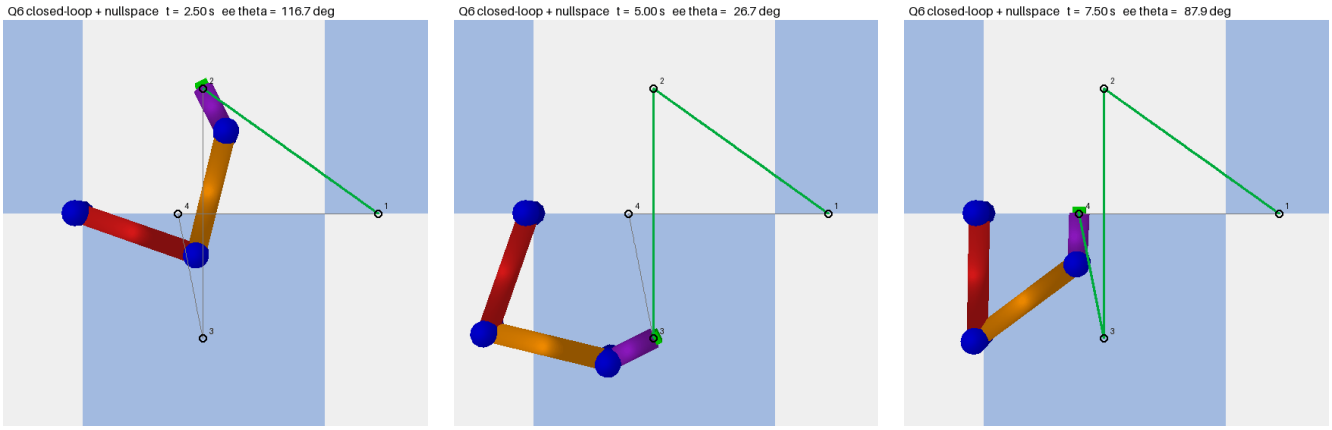


Figure 3: Q6 animation frames at $t = 2.5$, 5.0 and 7.5 s (left to right). The same locations are reached while orientation is free.

Animation: `q6_closed_loop.gif` (20 fps, 10.05 s). The maximum measured position error is below 2 μm. The nullspace identity is exact in the kinematic model; the simulation follows the prescribed path to this numerical accuracy.